Reg. No.	2	4	1	5	0	1	0	8	0

RAJALAKSHMI ENGINEERING COLLEGE

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Continuous Assessment Test-I [CAT -I]

Year : I
Semester : II
Branch : B.Tech AI&ML
Sub. Code : AI23231
Subject Name : Principles of Artificial Intelligence
QP Code : 011155

(Regulations 2023)
(Common to AI&DS)

Date: 16.04.2025 **Time: 120 Min** **Max. Marks: 75**

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	Define an "intelligent agent".	[CO1]	[A1]
2	Describe the difference between reactive and proactive agents.	[CO1]	[A2]
3	What is a heuristic function?	[CO1]	[A1]
4	Define model based agent. Give an example.	[CO2]	[A2]
5	Describe the difference between breadth-first search and depth-first search.	[CO2]	[A2]
6	What are the characteristics of an environment in AI?	[CO2]	[A1]
7	Explain the concept of "optimization problems" in the context of local search algorithms.	[CO2]	[A2]
8	Name a real-world application where A* search would be particularly effective.	[CO2]	[A1]
9	List the components of a well-defined problem.	[CO2]	[A1]
10	Define "constraint satisfaction problem" with an example.	[CO3]	[A2]

Part B: 5x11=55 Marks

11	a.	Discuss the applications and implications of using heuristic search strategies such as A* search in AI. Include a diagram to illustrate how A* search works.	[CO1]	[B1]

[OR]

b. Explain the various properties of environments in AI. How do these properties affect agent design? Provide suitable example. [CO1] [B1]

12 a. Illustrate with diagrams the concept of problem space in AI, explaining how different search algorithms navigate this space. [CO1] [B2]

[OR]

b. Describe the utility function in AI and its significance in utility-based agents' decision-making processes, using appropriate diagrams. [CO1] [B2]

13 a. Provide a detailed comparison of Greedy best-first search and A* search, including their applications and limitations. Include a diagram showing the state space traversal by each algorithm. [CO2] [B1]

[OR]

b. Explain the concept of depth-limited search and its application, with a flowchart representing its operation. [CO2] [B1]

14 a. Illustrate with diagrams the differences between local beam search and simulated annealing search, focusing on their mechanisms and typical use cases. [CO2] [B2]

[OR]

b. Discuss the principles and implementation of AO* search, including a diagram of the search tree it generates. [CO2] [B2]

15 a. Design an intelligent agent for a smart irrigation system. Use the PEAS framework and explain how the environment properties influence agent behaviour. [CO3] [B1]

[OR]

b. Analyze the application of constraint satisfaction algorithms in urban planning and infrastructure development. Provide a detailed explanation of how these algorithms help in the optimization of land use, traffic management, and utility distribution. Include diagrams or models to represent the problem-solving process. [CO3] [B2]

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Register No.: 231001203

Continuous Assessment-I		
Information Technology		
Degree: B. Tech	Semester: V	Year: III
Course code & Title:	AI23231 Principles of Artificial Intelligence	
Date: 25/08/2025		QP CODE: 101104

[Regulations: 2023]

Time: 02.00 hours Max. Marks: 75

Answer All Questions
Part A: 10 x 2 = 20 Marks

1	What is meant by the nature of the environment in AI?	[CO1]	[k2]
2	Name any two issues in the design of search programs.	[CO1]	[k1]
3	Which agent type is capable of improving its performance from past experiences?	[CO1]	[k2]
4	State one difference between goal-based and utility-based agents.	[CO1]	[k2]
5	Which search strategy searches both forward from the start and backward from the goal?	[CO2]	[k1]
6	Name two limitations of depth-limited search.	[CO2]	[k1]
7	List any two memory-bounded heuristic search algorithms.	[CO2]	[k1]
8	What is the main objective of local beam search?	[CO2]	[k2]
9	In constraint satisfaction problems, what does "arc consistency" mean?	[CO3]	[k2]
10	State one key difference between zero-sum games and non-zero-sum games.	[CO3]	[k2]

Part B: 5x11=55 Marks

11	a.	Explain AI techniques with a focus on the components: representation, search, and reasoning. Include a diagram showing the relationship among these components.	[CO1]	[k2]
		[OR]		
	b.	Describe the classification of environments in AI (fully/partially observable, deterministic/stochastic, etc.). Include a table and diagram illustrating each type.	[CO1]	[k2]

12	a.	Discuss learning agents in detail, explaining each component's role. Draw a labelled architecture diagram of a learning agent.	[CO1]	[k2]
		[OR]		
	b.	With a suitable example, define a problem in terms of state space search. Show the production rules and a state transition diagram.	[CO1]	[k3]
13	a.	Describe bidirectional search in detail with pseudocode. Illustrate the search process using an example.	[CO2]	[k3]
		[OR]		
	b.	Explain the comparison criteria for uniform search strategies, including time complexity, space complexity, completeness, and optimality. Present in tabular form.	[CO2]	[k2]
14	a.	Describe greedy best-first search with an example. Show node expansion order in a heuristic search diagram.	[CO2]	[k3]
		[OR]		
	b.	Explain the process and advantages of simulated annealing over hill climbing. Include a temperature vs. iteration graph.	[CO2]	[k3]
15	a.	Explain the working of alpha-beta pruning in minimax search with an example game tree diagram. Highlight the pruned branches.	[CO3]	[k3]
		[OR]		
	b.	Discuss the backtracking algorithm for solving CSPs with an example. Include a constraint graph and backtracking steps diagram.	[CO3]	[k3]

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Register No.: 231401001

Continuous Assessment-I		
Computer Science and Business Systems		
Degree: B. Tech.	Semester: V	Year: III
Course code & Title: CB23532	Artificial Intelligence	QP CODE: 141101
Date: 20/08/2025		

[Regulations: 2023]

Time: 02.00 hours

Max. Marks: 75

Answer All Questions
Part A: 10 x 2 = 20 Marks

1	Define an intelligent agent with an example.	[CO1]	[k1]
2	List any two differences between goal-based and utility-based agents.	[CO1]	[k2]
3	What is meant by AI technique? Give one example.	[CO1]	[k1]
4	Name any two limitations of the Tic-Tac-Toe problem as an AI model.	[CO1]	[k2]
5	Which data structure is used in breadth-first search?	[CO2]	[k1]
6	State one advantage and one disadvantage of depth-first search.	[CO2]	[k2]
7	What is the role of the evaluation function in A* search?	[CO2]	[k1]
8	List two applications of simulated annealing search.	[CO2]	[k2]
9	What is the purpose of the minimax algorithm in AI game playing?	[CO3]	[k2]
10	State the condition where alpha-beta pruning becomes most effective.	[CO3]	[k2]

Part B: 5x11=55 Marks

11	a.	Explain the structure of intelligent agents with a neat diagram. Describe the working principles of goal-based, utility-based, and learning agents with suitable examples.	[CO1]	[k3]
		[OR]		
	b.	Discuss the types of environments in which agents operate. Explain how environment properties influence the choice of agent architecture, supported by examples.	[CO1]	[k4]

12	a.	Identify major problems in AI such as the frame problem, combinatorial explosion, and brittleness. Explain how AI techniques help address these problems with real-world cases.	[CO1]	[k3]
		[OR]		
	b.	Model the Tic-Tac-Toe game as a state-space problem. Illustrate the game tree, branching factor, and winning strategy using diagrams.	[CO1]	[k3]
13	a.	Compare Breadth-First Search, Depth-First Search, Depth-Limited Search, and Bidirectional Search in terms of completeness, optimality, time complexity, and space complexity. Include a summary table and applications.	[CO2]	[k4]
		[OR]		
	b.	Explain the working of bidirectional search with a diagram. Analyze its advantages, limitations, and best-use scenarios.	[CO2]	[k4]
14	a.	Trace the evolution from Greedy Best-First Search to A* search. Discuss principles, evaluation functions, and factors affecting their performance, using an example problem.	[CO2]	[k4]
		[OR]		
	b.	Describe the process of simulated annealing search with a neat flow diagram. Discuss temperature control, acceptance probability, and an application in optimization.	[CO2]	[k3]
15	a.	Explain the minimax algorithm with a neat game tree diagram. Discuss how optimal decisions are derived in competitive games.	[CO3]	[k3]
		[OR]		
	b.	Illustrate alpha-beta pruning with an example game tree. Show how pruning reduces the number of evaluated nodes and improves efficiency.	[CO3]	[k4]